(21)

Figure 12.6.7

DISTANCE PROBLEMS INVOLVING PLANES

Next we will consider three basic "distance problems" in 3-space:

- Find the distance between a point and a plane.
- Find the distance between two parallel planes.
- Find the distance between two skew lines.

The three problems are related. If we can find the distance between a point and a plane, then we can find the distance between parallel planes by computing the distance between one of the planes and an arbitrary point P_0 in the other plane (Figure 12.6.8*a*). Moreover, we can find the distance between two skew lines by computing the distance between parallel planes containing them (Figure 12.6.8*b*).

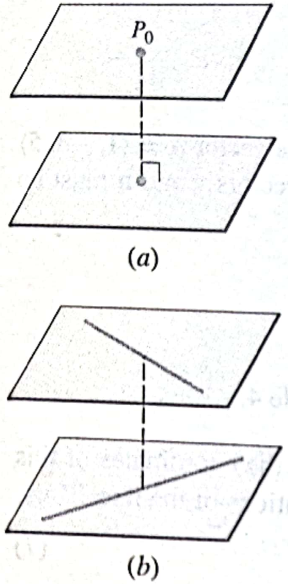


Figure 12.6.8

12.6.2 THEOREM. *The distance D between a point $P_0(x_0, y_0, z_0)$ and the plane $ax + by + cz + d = 0$ is*

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}} \tag{10}$$

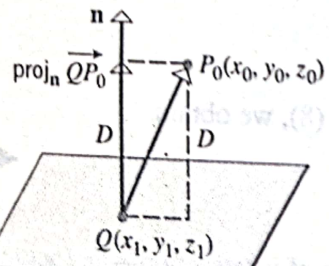


Figure 12.6.9

Proof. Let $Q(x_1, y_1, z_1)$ be any point in the plane, and position the normal $\mathbf{n} = \langle a, b, c\rangle$ so that its initial point is at Q. As illustrated in Figure 12.6.9, the distance D is equal to the length of the orthogonal projection of $\overrightarrow{QP_0}$ on $\mathbf{n}$. Thus, from (12) of Section 12.3,

$$D = \|\text{proj}_{\mathbf{n}}\overrightarrow{QP_0}\| = \left\|\frac{\overrightarrow{QP_0}\cdot\mathbf{n}}{\|\mathbf{n}\|^2}\mathbf{n}\right\| = \frac{|\overrightarrow{QP_0}\cdot\mathbf{n}|}{\|\mathbf{n}\|^2}\|\mathbf{n}\| = \frac{|\overrightarrow{QP_0}\cdot\mathbf{n}|}{\|\mathbf{n}\|}$$

But

$$\overrightarrow{QP_0} = \langle x_0 - x_1, y_0 - y_1, z_0 - z_1\rangle$$
$$\overrightarrow{QP_0}\cdot\mathbf{n} = a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)$$
$$\|\mathbf{n}\| = \sqrt{a^2 + b^2 + c^2}$$

Thus,

$$D = \frac{|a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)|}{\sqrt{a^2 + b^2 + c^2}} \tag{11}$$

Since the point $Q(x_1, y_1, z_1)$ lies in the plane, its coordinates satisfy the equation of the plane; that is,

$$ax_1 + by_1 + cz_1 + d = 0$$

or

$$d = -ax_1 - by_1 - cz_1$$

Combining this expression with (11) yields (10). ∎

Example 7 Find the distance D between the point $(1, -4, -3)$ and the plane

$$2x - 3y + 6z = -1$$

Solution. Formula (10) requires the plane to be rewritten in the form $ax+by+cz+d = 0$. Thus, we rewrite the equation of the given plane as

$$2x - 3y + 6z + 1 = 0$$

from which we obtain $a = 2, b = -3, c = 6$, and $d = 1$. Substituting these values and the coordinates of the given point in (10), we obtain

$$D = \frac{|(2)(1) + (-3)(-4) + 6(-3) + 1|}{\sqrt{2^2 + (-3)^2 + 6^2}} = \frac{|-3|}{7} = \frac{3}{7}$$ ◀

REMARK. See Exercise 48 for an analog of Formula (10) in 2-space that can be used to compute the distance between a point and a line.

Example 8 The planes

$$x + 2y - 2z = 3 \quad \text{and} \quad 2x + 4y - 4z = 7$$

are parallel since their normals, $\langle 1, 2, -2\rangle$ and $\langle 2, 4, -4\rangle$, are parallel vectors. Find the distance between these planes.

Solution. To find the distance D between the planes, we can select an arbitrary point in one of the planes and compute its distance to the other plane. By setting $y = z = 0$ in the equation $x + 2y - 2z = 3$, we obtain the point $P_0(3, 0, 0)$ in this plane. From (10), the distance from P_0 to the plane $2x + 4y - 4z = 7$ is

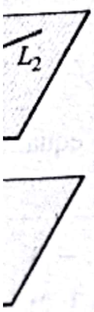
$$D = \frac{|(2)(3) + 4(0) + (-4)(0) - 7|}{\sqrt{2^2 + 4^2 + (-4)^2}} = \frac{1}{6}$$ ◀

Example 9 It was shown in Example 3 of Section 12.5 that the lines

$$L_1: x = 1 + 4t, \quad y = 5 - 4t, \quad z = -1 + 5t$$

$$L_2: x = 2 + 8t, \quad y = 4 - 3t, \quad z = 5 + t$$

are skew. Find the distance between them.



Solution. Let P_1 and P_2 denote parallel planes containing L_1 and L_2, respectively (Figure 12.6.10). To find the distance D between L_1 and L_2, we will calculate the distance from a point in P_1 to the plane P_2. Since L_1 lies in plane P_1, we can find a point in P_1 by finding a point on the line L_1; we can do this by substituting any convenient value of t in the parametric equations of L_1. The simplest choice is $t = 0$, which yields the point $Q_1(1, 5, -1)$.

Thus,

$$D = \frac{|a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)|}{\sqrt{a^2 + b^2 + c^2}} \tag{11}$$

Since the point $Q(x_1, y_1, z_1)$ lies in the plane, its coordinates satisfy the equation of the plane; that is,

$$ax_1 + by_1 + cz_1 + d = 0$$

or

$$d = -ax_1 - by_1 - cz_1$$

Combining this expression with (11) yields (10). ∎

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$$2x - 3y + 6z + 1 = 0$$

from which we obtain $a = 2$, $b = -3$, $c = 6$, and $d = 1$. Substituting these values and the coordinates of the given point in (10), we obtain

$$D = \frac{|(2)(1) + (-3)(-4) + 6(-3) + 1|}{\sqrt{2^2 + (-3)^2 + 6^2}} = \frac{|-3|}{7} = \frac{3}{7}$$ ◀

REMARK. See Exercise 48 for an analog of Formula (10) in 2-space that can be used to compute the distance between a point and a line.

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are parallel since their normals, $\langle 1, 2, -2\rangle$ and $\langle 2, 4, -4\rangle$, are parallel vectors. Find the distance between these planes.

Solution. To find the distance D between the planes, we can select an arbitrary point in one of the planes and compute its distance to the other plane. By setting $y = z = 0$ in the equation $x + 2y - 2z = 3$, we obtain the point $P_0(3, 0, 0)$ in this plane. From (10), the distance from P_0 to the plane $2x + 4y - 4z = 7$ is

$$D = \frac{|(2)(3) + 4(0) + (-4)(0) - 7|}{\sqrt{2^2 + 4^2 + (-4)^2}} = \frac{1}{6}$$ ◀

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$$L_2: x = 2 + 8t, \quad y = 4 - 3t, \quad z = 5 + t$$

are skew. Find the distance between them.

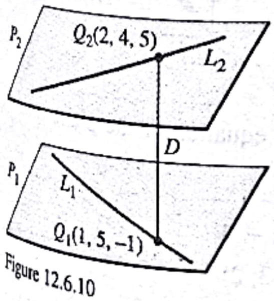


Figure 12.6.10

Solution. Let P_1 and P_2 denote parallel planes containing L_1 and L_2, respectively (Figure 12.6.10). To find the distance D between L_1 and L_2, we will calculate the distance from a point in P_1 to the plane P_2. Since L_1 lies in plane P_1, we can find a point in P_1 by finding a point on the line L_1; we can do this by substituting any convenient value of t in the parametric equations of L_1. The simplest choice is $t = 0$, which yields the point $Q_1(1, 5, -1)$.

The next step is to find an equation for the plane P_2. For this purpose, observe that the vector $\mathbf{u}_1 = \langle 4, -4, 5\rangle$ is parallel to line L_1, and therefore also parallel to planes P_1 and P_2. Similarly, $\mathbf{u}_2 = \langle 8, -3, 1\rangle$ is parallel to L_2 and hence parallel to P_1 and P_2. Therefore, the cross product

$$\mathbf{n} = \mathbf{u}_1 \times \mathbf{u}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -4 & 5 \\ 8 & -3 & 1 \end{vmatrix} = 11\mathbf{i} + 36\mathbf{j} + 20\mathbf{k}$$

is normal to both P_1 and P_2. Using this normal and the point $Q_2(2, 4, 5)$ found by setting $t = 0$ in the equations of L_2, we obtain an equation for P_2:

$$11(x-2) + 36(y-4) + 20(z-5) = 0$$

or

$$11x + 36y + 20z - 266 = 0$$

The distance between $Q_1(1, 5, -1)$ and this plane is

$$D = \frac{|(11)(1) + (36)(5) + (20)(-1) - 266|}{\sqrt{11^2 + 36^2 + 20^2}} = \frac{95}{\sqrt{1817}}$$

which is also the distance between L_1 and L_2. ◀

EXERCISE SET 12.6

...of the plane indicated in

we can regard the line as the line of intersection of the two planes

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} \qquad \text{and} \qquad \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

EXAMPLE 8 Find a formula for the distance D from a point $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$.

SOLUTION Let $P_0(x_0, y_0, z_0)$ be any point in the given plane and let $\mathbf{b}$ be the vector corresponding to $\overrightarrow{P_0P_1}$. Then

$$\mathbf{b} = \langle x_1 - x_0, y_1 - y_0, z_1 - z_0 \rangle$$

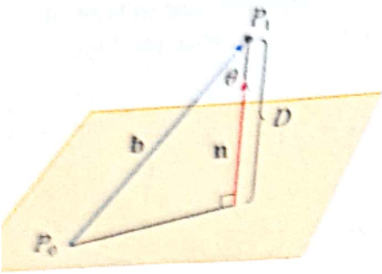


FIGURE 12

From Figure 12 you can see that the distance D from P_1 to the plane is equal to the absolute value of the scalar projection of $\mathbf{b}$ onto the normal vector $\mathbf{n} = \langle a, b, c \rangle$. (See Section 9.3.) Thus

$$D = |\text{comp}_{\mathbf{n}}\,\mathbf{b}| = \frac{|\mathbf{n} \cdot \mathbf{b}|}{|\mathbf{n}|}$$

$$= \frac{|a(x_1 - x_0) + b(y_1 - y_0) + c(z_1 - z_0)|}{\sqrt{a^2 + b^2 + c^2}}$$

$$= \frac{|(ax_1 + by_1 + cz_1) - (ax_0 + by_0 + cz_0)|}{\sqrt{a^2 + b^2 + c^2}}$$

Since P_0 lies in the plane, its coordinates satisfy the equation of the plane and so we have $ax_0 + by_0 + cz_0 + d = 0$. Thus, the formula for D can be written as

8 $$\boxed{D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}}$$

EXAMPLE 9 Find the distance between the parallel planes $10x + 2y - 2z = 5$ and $5x + y - z = 1$.

SOLUTION First we note that the planes are parallel because their normal vectors $\langle 10, 2, -2 \rangle$ and $\langle 5, 1, -1 \rangle$ are parallel. To find the distance D between the planes, we choose any point on one plane and calculate its distance to the other plane. In particular, if we put $y = z = 0$ in the equation of the first plane, we get $10x = 5$ and so $(\frac{1}{2}, 0, 0)$ is a point in this plane. By Formula 8, the distance between $(\frac{1}{2}, 0, 0)$ and the plane $5x + y - z - 1 = 0$ is

$$D = \frac{|5(\frac{1}{2}) + 1(0) - 1(0) - 1|}{\sqrt{5^2 + 1^2 + (-1)^2}} = \frac{\frac{3}{2}}{3\sqrt{3}} = \frac{\sqrt{3}}{6}$$

So the distance between the planes is $\sqrt{3}/6$.

EXAMPLE 10 In Example 3 we showed that the lines

$$L_1: \quad x = 1 + t \qquad y = -2 + 3t \qquad z = 4 - t$$

$$L_2: \quad x = 2s \qquad y = 3 + s \qquad z = -3 + 4s$$

are skew. Find the distance between them.

planes P_1 and P_2. ... L_1 and L_2 are skew, they can be viewed as lying on two planes P_1 and P_2. The distance between L_1 and L_2 is the same as the distance between P_1 and P_2, which can be computed as in Example 9. The common normal vector to both planes must be orthogonal to both $\mathbf{v}_1 = \langle 1, 3, -1 \rangle$ (the direction of L_1) and $\mathbf{v}_2 = \langle 2, 1, 4 \rangle$ (the direction of L_2). So a normal vector is

$$\mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & -1 \\ 2 & 1 & 4 \end{vmatrix} = 13\mathbf{i} - 6\mathbf{j} - 5\mathbf{k}$$

If we put $s = 0$ in the equations of L_2, we get the point $(0, 3, -3)$ on L_2 and so an equation for P_2 is

$$13(x - 0) - 6(y - 3) - 5(z + 3) = 0 \qquad \text{or} \qquad 13x - 6y - 5z + 3 = 0$$

If we now set $t = 0$ in the equations for L_1, we get the point $(1, -2, 4)$ on P_1. So the distance between L_1 and L_2 is the same as the distance from $(1, -2, 4)$ to $13x - 6y + 5z + 3 = 0$. By Formula 8, this distance is

$$D = \frac{|13(1) - 6(-2) - 5(4) + 3|}{\sqrt{13^2 + (-6)^2 + (-5)^2}} = \frac{8}{\sqrt{230}} \approx 0.53$$

Exercises

...etermine whether each statement is true or false.

...) Two lines parallel to a third line are parallel.

...) Two lines perpendicular to a third line are parallel.

...) Two planes parallel to a third plane are parallel.

...) Two planes perpendicular to a third plane are parallel.

...) Two lines parallel to a plane are parallel.

...) Two lines perpendicular to a plane are parallel.

...) Two planes parallel to a line are parallel.

...) Two planes perpendicular to a line are parallel.

...) Two planes either intersect or are parallel.

...) Two lines either intersect or are parallel.

...) A plane and a line either intersect or are parallel.

... Find a vector equation and parametric equations for ...ne.

...he line through the point $(1, 0, -3)$ and parallel to the ...ector $2\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$

...he line through the point $(-2, 4, 10)$ and parallel to the ...ector $\langle 3, 1, -8 \rangle$

...he line through the origin and parallel to the line $x = 2t$, ...$= 1 - t$, $z = 4 + 3t$

...he line through the point $(1, 0, 6)$ and perpendicular to the ...plane $x + 3y + z = 5$

6–10 ■ Find parametric equations and symmetric equations for the line.

6. The line through the origin and the point $(1, 2, 3)$

7. The line through the points $(3, 1, -1)$ and $(3, 2, -6)$

8. The line through the points $(-1, 0, 5)$ and $(4, -3, 3)$

9. The line through the points $(0, \frac{1}{2}, 1)$ and $(2, 1, -3)$

10. The line of intersection of the planes $x + y + z = 1$ and $x + z = 0$

11. Show that the line through the points $(2, -1, -5)$ and $(8, 8, 7)$ is parallel to the line through the points $(4, 2, -6)$ and $(8, 8, 2)$.

12. Show that the line through the points $(0, 1, 1)$ and $(1, -1, 6$...) is perpendicular to the line through the points $(-4, 2, 1)$ and $(-1, 6, 2)$.

13. (a) Find symmetric equations for the line that passes through the point $(0, 2, -1)$ and is parallel to the line with parametric equations $x = 1 + 2t$, $y = 3t$, $z = 5 - 7t$.

(b) Find the points in which the required line in part (a) intersects the coordinate planes.

14. (a) Find parametric equations for the line through (5, 1, 0) that is perpendicular to the plane $2x - y + z = 1$.
(b) In what points does this line intersect the coordinate planes?

15–18 ■ Determine whether the lines L_1 and L_2 are parallel, skew, or intersecting. If they intersect, find the point of intersection.

15. L_1: $\dfrac{x-4}{2} = \dfrac{y+5}{4} = \dfrac{z-1}{-3}$

L_2: $\dfrac{x-2}{1} = \dfrac{y+1}{3} = \dfrac{z}{2}$

16. L_1: $\dfrac{x-1}{2} = \dfrac{y}{1} = \dfrac{z-1}{4}$,

L_2: $\dfrac{x}{1} = \dfrac{y+2}{2} = \dfrac{z+2}{3}$

17. L_1: $x = -6t$, $y = 1 + 9t$, $z = -3t$
L_2: $x = 1 + 2s$, $y = 4 - 3s$, $z = s$

18. L_1: $x = 1 + t$, $y = 2 - t$, $z = 3t$
L_2: $x = 2 - s$, $y = 1 + 2s$, $z = 4 + s$

19–28 ■ Find an equation of the plane.

19. The plane through the point (6, 3, 2) and perpendicular to the vector $\langle -2, 1, 5\rangle$

20. The plane through the point (4, 0, −3) and with normal vector $\mathbf{j} + 2\mathbf{k}$

21. The plane through the origin and parallel to the plane $2x - y + 3z = 1$

22. The plane that contains the line $x = 3 + 2t$, $y = t$, $z = 8 - t$ and is parallel to the plane $2x + 4y + 8z = 17$

23. The plane through the points (0, 1, 1), (1, 0, 1), and (1, 1, 0)

24. The plane through the origin and the points (2, −4, 6) and (5, 1, 3)

25. The plane that passes through the point (6, 0, −2) and contains the line $x = 4 - 2t$, $y = 3 + 5t$, $z = 7 + 4t$

26. The plane that passes through the point (1, −1, 1) and contains the line with symmetric equations $x = 2y = 3z$

27. The plane that passes through the point (−1, 2, 1) and contains the line of intersection of the planes $x + y - z = 2$ and $2x - y + 3z = 1$

28. The plane that passes through the line of intersection of the planes $x - z = 1$ and $y + 2z = 3$ and is perpendicular to the plane $x + y - 2z = 1$

29–30 ■ Find the point at which the line intersects the given plane.

29. $x = 1 + 2t$, $y = -1$, $z = t$; $2x + y - z + 5 = 0$

30. $x = 1 - t$, $y = t$, $z = 1 + t$; $z = 1 - 2x + y$

31–34 ■ Determine whether the planes are parallel, perpendicular, or neither. If neither, find the angle between them.

31. $x + z = 1$, $y + z = 1$

32. $-8x - 6y + 2z = 1$, $z = 4x + 3y$

33. $x + 4y - 3z = 1$, $-3x + 6y + 7z = 0$

34. $2x + 2y - z = 4$, $6x - 3y + 2z = 5$

35. (a) Find symmetric equations for the line of intersection of the planes $x + y - z = 2$ and $3x - 4y + 5z = 6$.
(b) Find the angle between these planes.

36. Find an equation for the plane consisting of all points that are equidistant from the points (−4, 2, 1) and (2, −4, 3).

37. Find an equation of the plane with x-intercept a, y-intercept b, and z-intercept c.

38. (a) Find the point at which the given lines intersect:

$$\mathbf{r} = \langle 1, 1, 0\rangle + t\langle 1, -1, 2\rangle$$

and $$\mathbf{r} = \langle 2, 0, 2\rangle + s\langle -1, 1, 0\rangle$$

(b) Find an equation of the plane that contains these lines.

39. Find parametric equations for the line through the point (0, 1, 2) that is parallel to the plane $x + y + z = 2$ and perpendicular to the line $x = 1 + t$, $y = 1 - t$, $z = 2t$.

40. Find parametric equations for the line through the point (0, 1, 2) that is perpendicular to the line $x = 1 + t$, $y = 1 - t$, $z = 2t$ and intersects this line.

41. Which of the following four planes are parallel? Are any of them identical?

P_1: $4x - 2y + 6z = 3$ P_2: $4x - 2y - 2z = 6$

P_3: $-6x + 3y - 9z = 5$ P_4: $z = 2x - y - 3$

42. Which of the following four lines are parallel? Are any of them identical?

L_1: $x = 1 + t$, $y = t$, $z = 2 - 5t$

L_2: $x + 1 = y - 2 = 1 - z$

L_3: $x = 1 + t$, $y = 4 + t$, $z = 1 - t$

L_4: $\mathbf{r} = \langle 2, 1, -3\rangle + t\langle 2, 2, -10\rangle$

43–44 ■ Use the formula in Exercise 27 in Section 9.4 to find the distance from the point to the given line.

43. (1, 2, 3); $x = 2 + t$, $y = 2 - 3t$, $z = 5t$

44. (1, 0, −1); $x = 5 - t$, $y = 3t$, $z = 1 + 2t$

45–46 ■ Find the distance from the point to the given plane.

45. (2, 8, 5), $x - 2y - 2z = 1$

46. (3, −2, 7), $4x - 6y + z = 5$

-48 ▪ Find the distance between the given parallel planes.

$z = x + 2y + 1, \quad 3x + 6y - 3z = 4$

$3x + 6y - 9z = 4, \quad x + 2y - 3z = 1$

Show that the distance between the parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is

$$D = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$$

Find equations of the planes that are parallel to the plane $x + 2y - 2z = 1$ and two units away from it.

Show that the lines with symmetric equations $x = y = z$ and $x + 1 = y/2 = z/3$ are skew, and find the distance between these lines.

52. Find the distance between the skew lines with parametric equations $x = 1 + t$, $y = 1 + 6t$, $z = 2t$, and $x = 1 + 2s$, $y = 5 + 15s$, $z = -2 + 6s$.

53. If a, b, and c are not all 0, show that the equation $ax + by + cz + d = 0$ represents a plane and $\langle a, b, c \rangle$ is a normal vector to the plane.

Hint: Suppose $a \neq 0$ and rewrite the equation in the form

$$a\left(x + \frac{d}{a}\right) + b(y - 0) + c(z - 0) = 0$$

54. Give a geometric description of each family of planes.

(a) $x + y + z = c$

(b) $x + y + cz = 1$

(c) $y \cos \theta + z \sin \theta = 1$

40. $(0, 1, 5)$; $3x + 6y - 2z - 5 = 0$

In Exercises 41 and 42, find the distance between the given parallel planes.

41. $-2x + y + z = 0$
$6x - 3y - 3z - 5 = 0$

42. $x + y + z = 1$
$x + y + z = -1$

In Exercises 43 and 44, find the distance between the given skew lines.

43. $x = 1 + 7t,\ y = 3 + t,\ z = 5 - 3t$
$x = 4 - t,\ y = 6,\ z = 7 + 2t$

44. $x = 3 - t,\ y = 4 + 4t,\ z = 1 + 2t$
$x = t,\ y = 3,\ z = 2t$

45. Find an equation of the sphere with center $(2, 1, -3)$ that is tangent to the plane $x - 3y + 2z = 4$.

46. Locate the point of intersection of the plane $2x + y - z = 0$ and the line through $(3, 1, 0)$ that is perpendicular to the plane.

47. Show that the line $x = -1 + t$, $y = 3 + 2t$, $z = -t$ and the plane $2x - 2y - 2z + 3 = 0$ are parallel, and find the distance between them.

48. Formulas (1), (2), (3), (5), and (10), which apply to planes in 3-space, have analogs for lines in 2-space.

(a) Draw an analog of Figure 12.6.3 in 2-space to illustrate that the equation of the line that passes through the point $P(x_0, y_0)$ and is perpendicular to the vector $\mathbf{n} = \langle a, b\rangle$ can be expressed as

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$$

where $\mathbf{r} = \langle x, y\rangle$ and $\mathbf{r}_0 = \langle x_0, y_0\rangle$.

(b) Show that the vector equation in part (a) can be expressed as

$$a(x - x_0) + b(y - y_0) = 0$$

This is called the ***point-normal form of a line***.

(c) Using the proof of Theorem 12.6.1 as a guide, show that if a and b are not both zero, then the graph of the equation

$$ax + by + c = 0$$

is a line that has $\mathbf{n} = \langle a, b\rangle$ as a normal.

(d) Using the proof of Theorem 12.6.2 as a guide, show that the distance D between a point $P(x_0, y_0)$ and the line $ax + by + c = 0$ is

$$D = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

49. Use the formula in part (d) of Exercise 48 to find the distance between the point $P(-3, 5)$ and the line $y = -2x + 1$.

50. (a) Show that the distance D between parallel planes

$$ax + by + cz + d_1 = 0$$
$$ax + by + cz + d_2 = 0$$

is

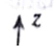
$$D = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$$

(b) Use the formula in part (a) to solve Exercise 41.

12.7 QUADRIC SURFACES

In this section we will study an important class of surfaces that are the three-dimensional analogs of the conic sections.

TRACES OF SURFACES



Although the general shape of a curve in 2-space can be obtained by plotting points, this method is not usually helpful for surfaces in 3-space because too many points are required. It is more common to build